

Documentacion Tarea 0: Instalacion y ejecucion de xv6

Grupo quillota-quilpué

1. Pasos de la instalacion

- Fork del repo oficial del mit /xv6-riscv
- Clonacion del repo con 'git clone' y creación de la nueva rama con 'git checkout -b ...'

```
ignac@LAPTOP-TM6C6NH5:~$ git clone https://github.com/ignacioosoto/xv6-riscv
Cloning into 'xv6-riscv'...
remote: Enumerating objects: 7788, done.
remote: Counting objects: 100% (279/279), done.
remote: Compressing objects: 100% (99/99), done.
remote: Total 7788 (delta 215), reused 180 (delta 180), pack-reused 7509 (from 4)
Receiving objects: 100% (7788/7788), 17.50 MiB | 1.68 MiB/s, done.
Resolving deltas: 100% (4114/4114), done.
ignac@LAPTOP-TM6C6NH5:~$ cd xv6-riscv
ignac@LAPTOP-TM6C6NH5:~/xv6-riscv$ git checkout -b users/ignac/ignacioosoto
Switched to a new branch 'users/ignac/ignacioosoto'
```

c. Instalacion de dependencias

```
ignac@LAPTOP-TM6C6NH5:~/xv6-riscv$ sudo apt update
sudo apt install gcc-riscv64-unknown-elf binutils-riscv64-unknown-elf gdb-multiarch
[sudo] password for ignac:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libipt2 libsource-highlight-common libsource-highlight4t64
Suggested packages:
  gdb-doc gdbserver
The following NEW packages will be installed:
  binutils-riscv64-unknown-elf gcc-riscv64-unknown-elf gdb gdb-multiarch libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libipt2 libsource-highlight-common
  libsource-highlight4t64
0 upgraded, 11 newly installed, 0 to remove and 0 not upgraded.
Need to get 68.6 MB of archives.
After this operation, 310 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod-common all 0.190-1.1ubuntu0.1 [14.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 binutils-riscv64-unknown-elf amd64 2.42-1ubuntu1+6 [2997 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 gcc-riscv64-unknown-elf amd64 13.2.0-11ubuntu1+2 [48.4 MB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libbabeltrace1 amd64 1.5.11-3build3 [164 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod1t64 amd64 0.190-1.1ubuntu0.1 [17.1 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 libipt2 amd64 2.0.6-1build1 [45.7 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight-common all 3.1.9-4.3build1 [64.2 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build1 [258 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 gdb amd64 15.0.50.20240403-0ubuntu1 [4010 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 gdb-multiarch amd64 15.0.50.20240403-0ubuntu1 [5152 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libc6-dbg amd64 2.39-0ubuntu8.5 [7458 kB]
Fetched 68.6 MB in 2min 16s (503 kB/s)
```

- d. Compilacion 'make' y ejecución con 'make qemu'

```
ignac@LAPTOP-TM6C6NH5:~/xv6-riscv$ make
make qemu

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$
```

2. Verificación de que funciona el sistema operativo con comandos básicos

- a. 'ls':

```
hart 2 starting
hart 1 starting
init: starting sh
$ ls
.          1 1 1024
..         1 1 1024
README    2 2 2425
cat        2 3 36504
echo       2 4 35360
forktest   2 5 17080
grep       2 6 39944
init       2 7 35808
kill       2 8 35280
ln         2 9 35096
ls         2 10 38640
mkdir      2 11 35344
rm         2 12 35328
sh         2 13 58344
stressfs   2 14 36208
usertests  2 15 186248
grind      2 16 51736
wc         2 17 37488
zombie     2 18 34680
logstress  2 19 37280
forphan    2 20 36104
dorphan    2 21 35544
console    3 22 0
$
```

- b. 'echo hola xv6':

```
$ echo HOLA XV6
HOLA XV6
$
```

c. 'cat README':

```
$ cat README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6).  xv6 loosely follows the structure and style of v6,
but is implemented for a modern RISC-V multiprocessor using ANSI C.

ACKNOWLEDGMENTS

xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
to Peer Communications; ISBN: 1-57398-013-7; 1st edition (June 14,
2000)).  See also https://pdos.csail.mit.edu/6.1810/, which provides
pointers to on-line resources for v6.

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locking), Cliff Frey (MP), Xiao Yu (MP), Nickolai Zeldovich, and Austin
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ERROR REPORTS

Please send errors and suggestions to Frans Kaashoek and Robert Morris
(kaashoek,rtm@mit.edu).  The main purpose of xv6 is as a teaching
operating system for MIT's 6.1810, so we are more interested in
simplifications and clarifications than new features.

BUILDING AND RUNNING XV6

You will need a RISC-V "newlib" tool chain from
https://github.com/riscv/riscv-gnu-toolchain, and qemu compiled for
```